

# 10.7.26

AI25BTECH11017-BALU

## Question:

- 1) The focal chord to  $y^2 = 16x$  is tangent to  $(x - 6)^2 + y^2 = 2$ , then the possible values of the slope of this chord are: (2003)
- a)  $-1, 1$
  - b)  $-2, 2$
  - c)  $-2, -\frac{1}{2}$
  - d)  $2, -\frac{1}{2}$

## Solution:

Let us solve the given equation theoretically and then verify the solution computationally. According to the question, The parabola is

$$y^2 = 16x. \quad (1.1)$$

In quadratic form, this can be written as

$$\mathbf{x}^T \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} -8 \\ 0 \end{pmatrix}^T \mathbf{x} = 0. \quad (1.2)$$

The circle is

$$(x - 6)^2 + y^2 = 2 \Rightarrow x^2 + y^2 - 12x + 34 = 0, \quad (1.3)$$

which can be expressed as

$$\mathbf{x}^T \mathbf{I} \mathbf{x} + 2 \begin{pmatrix} -6 \\ 0 \end{pmatrix}^T \mathbf{x} + 34 = 0. \quad (1.4)$$

Thus,

$$\mathbf{V} = \mathbf{I}, \quad \mathbf{u} = \begin{pmatrix} -6 \\ 0 \end{pmatrix}, \quad f = 34. \quad (1.5)$$

The focus of the parabola is

$$\mathbf{F} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}. \quad (1.6)$$

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*Equation of focal chord*

Any line through  $\mathbf{F}$  with slope  $m$  can be written as

$$y = m(x - 4). \quad (1.7)$$

In normal form, this becomes

$$\mathbf{n}^T \mathbf{x} = c, \quad (1.8)$$

where

$$\mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix}, \quad c = \mathbf{n}^T \mathbf{F} = -4m. \quad (1.9)$$

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*Tangency condition*

For a conic  $\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0$ , the line  $\mathbf{n}^T \mathbf{x} = c$  is tangent if

$$(\mathbf{n}^T \mathbf{C} - c)^2 = (\mathbf{n}^T \mathbf{V}^{-1} \mathbf{n})(\mathbf{u}^T \mathbf{V}^{-1} \mathbf{u} - f), \quad (1.10)$$

where  $\mathbf{C} = -\mathbf{V}^{-1} \mathbf{u}$  is the centre.

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*Applying to circle*

Here,

$$\mathbf{C} = -\mathbf{V}^{-1} \mathbf{u} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}, \quad \mathbf{u}^T \mathbf{V}^{-1} \mathbf{u} - f = 36 - 34 = 2. \quad (1.11)$$

So,

$$(\mathbf{n}^T \mathbf{C} - c)^2 = (\mathbf{n}^T \mathbf{n})(2). \quad (1.12)$$

Now,

$$\mathbf{n}^T \mathbf{C} = \begin{pmatrix} -m & 1 \end{pmatrix} \begin{pmatrix} 6 \\ 0 \end{pmatrix} = -6m, \quad (1.13)$$

and  $c = -4m$ .

Thus,

$$(-6m - (-4m))^2 = 2(m^2 + 1). \quad (1.14)$$

$$(-2m)^2 = 2(m^2 + 1). \quad (1.15)$$

$$4m^2 = 2m^2 + 2 \Rightarrow m^2 = 1. \quad (1.16)$$


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*Result*

$$m = \pm 1. \quad (1.17)$$

Hence, the slopes of the focal chord are

$$\boxed{-1 \text{ and } 1}.$$

(1.18)

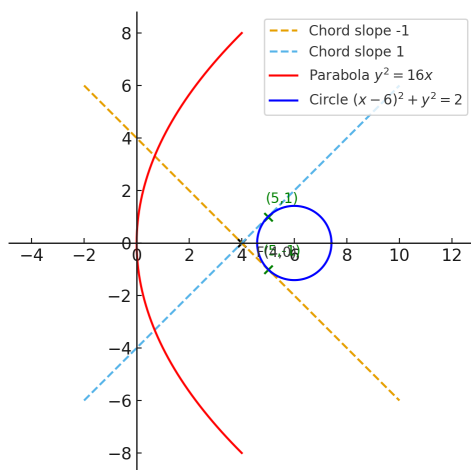


Fig. 1.1